Flexbox property

- ✓ **Flexbox** (short for *Flexible Box Layout*) is a **modern CSS layout model** that provides an efficient way to align, distribute, and space items inside a container even when their size is unknown or dynamic.
- ✓ It's designed to make it easier to create **responsive layouts**, where elements adjust automatically to different screen sizes and available space.

Property	Description	Possible Values	Syntax / Example
display: flex	Defines a flex container to arrange items flexibly.	flex, inline-flex	div { display: flex; }
flex-direction	Defines the direction of flex items.	row (default), row-reverse, column, column-reverse	flex-direction: row;
justify-content	Aligns items horizontally (along the main axis).	flex-start, flex-end, center, space-between, space-around, space-evenly	justify-content: space-between;
align-items	Aligns items vertically (along the cross axis).	stretch (default), flex-start, flex-end, center	align-items: center;
flex-wrap	Determines whether flex items wrap onto multiple lines.	nowrap (default), wrap, wrap-reverse	flex-wrap: wrap;
gap	Defines the space between flex items.	Any CSS length unit (px, em, %)	gap: 15px;
order	Specifies the display order of flex items.	Integer values (0 default, can be positive or negative)	order: 2;
flex *(shorthand)*	Sets how an item grows, shrinks, and defines its base size.	flex: grow shrink basis; e.g., flex: 1 0 100px;	flex: 1; or flex: 1 0 200px;

Flex Example

```
<head>
<style>
.container {
  display: flex;              /* Creates a flex container. Change values and check effects */
  flex-direction: row;        /* Arranges items in a row (default). Change values and check effects */
  justify-content: space-between; /* Spaces items horizontally. Change values and check effects */
  align-items: center;        /* Aligns items vertically in the center. Change values and check effects */
  flex-wrap: wrap;            /* Wraps items to next line if space runs out */
  gap: 15px;                  /* Adds space between items */
  background-color: lightgray;
  padding: 15px;
}
.item {
  background-color: steelblue;
  color: white;
  padding: 20px;
  text-align: center;
  border-radius: 5px;
  width: 200px;
```

```
}
.item:nth-child(2) {
  order: 1;            /* This will move the item to appear after others. Change values and check effects(POSITIVE,NEGATIVE,0) */
/*  flex: 1 1 100px;  */     /* Remove comments from this line to check effect. flex-grow: 1, flex-shrink: 1, flex-basis: 100px */
  background-color: orange;
}
</style>
</head>
<body>
<div class="container">
  <div class="item">Item 1</div>
  <div class="item">Item 2 (Order 1)</div>
  <div class="item">Item 3</div>
</div>
</body>
```



Property	Description	In This Example
display: flex	Turns the container into a flexbox container.	.container becomes a flex container, and its child .item elements become flex items.
flex-direction: row	Arranges flex items horizontally (left to right).	All .item boxes appear side by side.
justify-content: space-between	Distributes items evenly with space between them.	The first and last item stick to edges; middle items get even gaps.
align-items: center	Aligns items vertically along the cross-axis.	Centers items vertically within the container's height.
flex-wrap: wrap	Allows items to wrap to a new line if there's not enough space.	If you resize the browser, items will move to the next row.
gap: 15px	Adds space between flex items.	Creates uniform spacing (15px) between all .items.
order	Defines the visual order of items (default = 0).	order: 1 makes Item 2 appear **after** Items 1 and 3.
flex: 1 1 100px	Shorthand for flex-grow, flex-shrink, and flex-basis.	Item 2 starts at 100px wide, can grow and shrink with available space.

Normal Flow: Item 1 | Item 2 | Item 3
After order: 1 on Item 2: It moves after the other two → Item 1 | Item 3 | Item 2
Resize the browser window: Item 2 (with flex: 1 1 100px) will stretch or shrink smoothly as the container grows or shrinks.